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Evaluating Temporal Dimensionality Reduction Methods for ERP-Structured EEG Data

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Motivation

EEG data is:

- High-Dimensional
- Complex
- Ecologically Valid Designs



- **Exploratory analysis helps to** uncover meaningful patterns
- Achieved by Dimensionality Reduction

Research Questions

1. Do time-aware dimensionality reduction methods (BCNE/T-PHATE) preserve the temporal structure of ERP-related EEG data better than time-agnostic methods (PCA, t-SNE, UMAP, PHATE)?
2. Which metrics best evaluate temporal preservation in dimensionality reduction?

Hypotheses

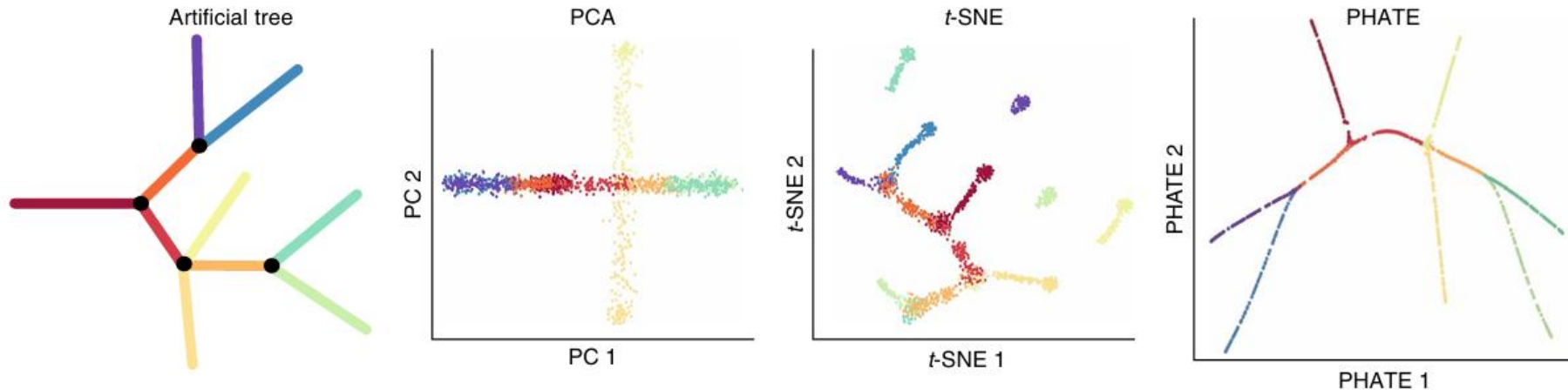
Time-aware methods will significantly outperform time-agnostic methods in the following:

- Preservation of the ground-truth structure of EEG/ERP data
- Representation of the continuous ERP component trajectory (P100–N170–P300–N400) and its divergence across conditions.

Dimensionality Reduction Methods Used

Method	Local Structure	Global Structure	Temporal Structure
PCA	-	+	-
t-SNE	+	-	-
UMAP	+	+	-
PHATE	+	+	-
T-PHATE	+	+	+
BCNE	+	+	+

PHATE: Potential of heat diffusion for affinity-based transition embedding



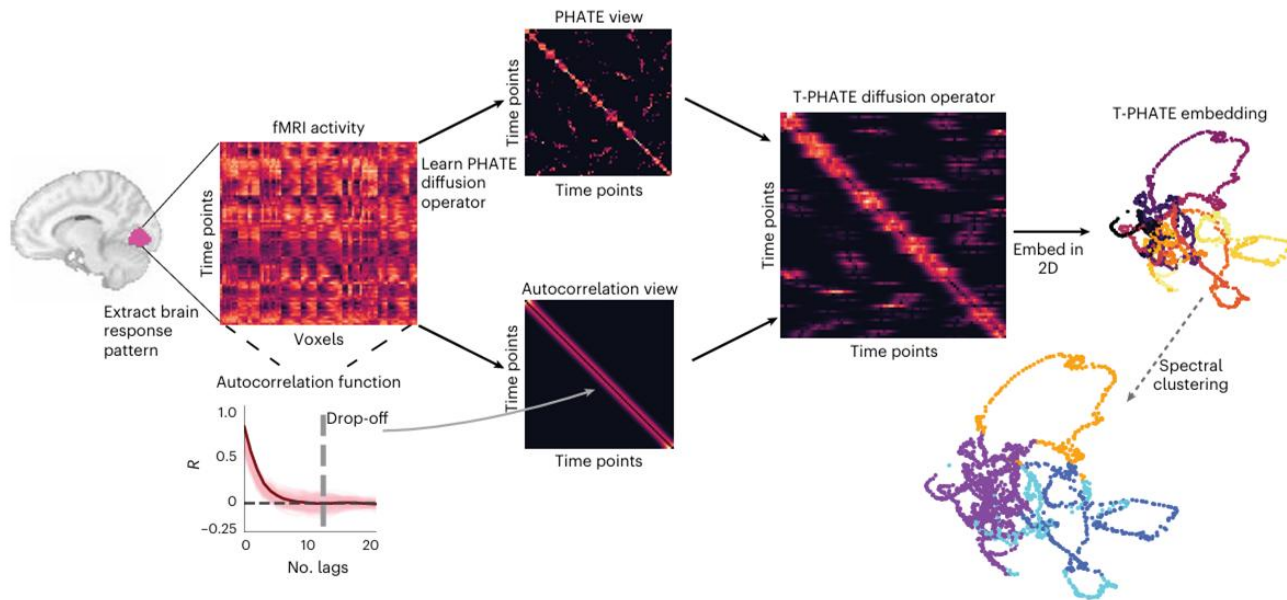
Core idea:

1. Calculate similarity between local data points using α decay kernel

2. Calculate distances between data points globally: Creates diffusion maps

3. Map distance information into low dimensions

TPHATE: Temporal PHATE

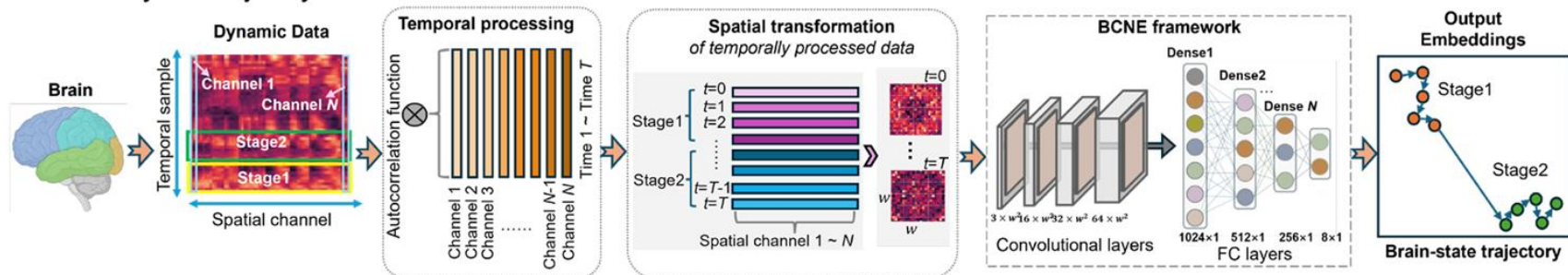


Temporal PHATE : PHATE + Time Autocorrelation = Diffusion operator

To understand both what the brain is doing and when it is doing it.

Brain-dynamic Convolutional-Network-based Embedding: Unsupervised learning

b. BCNE dynamic trajectory



Compute which time points correlate with others

Arrange similar channels in grid 6x6 for 32 channels

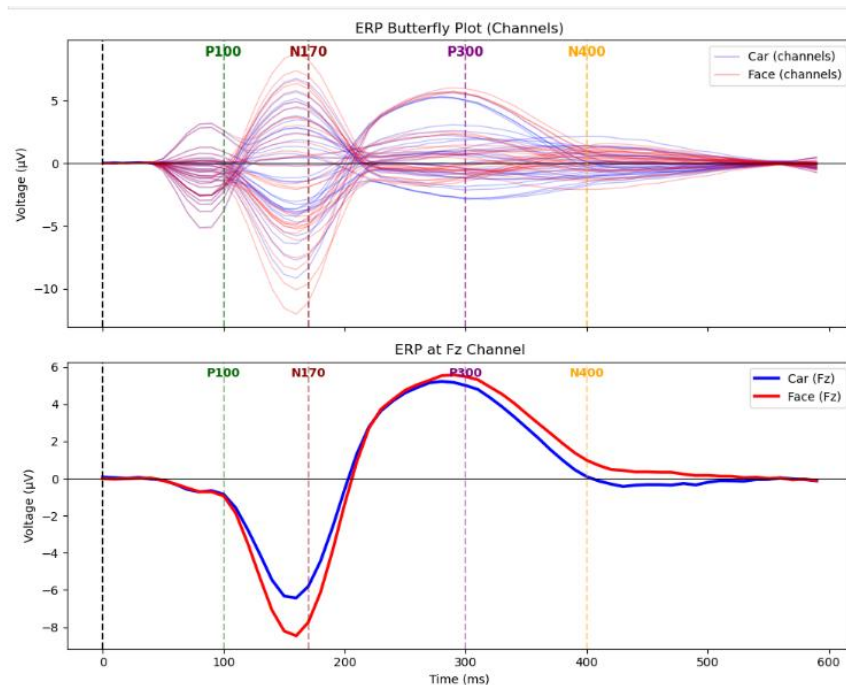
Recursive training:
Train 4 times.
Each iteration learns deeper features.

Unsupervised
2D embedding
shows condition separation

Dataset to be analyzed

Data simulated using UnfoldSim in Julia

- Conditions: ['car', 'face']
- Continuous: 10 levels from -5 to 5
[-5.0, -3.888, -2.7777, -1.66....., 5.0]
- Total trials: 2000
- Each trial lasted for 600 ms.
- Timepoints per trial: 60ms
- ERP components:
 - P100 (baseline, no effects)
 - N170 (condition effect)
 - P300 (continuous effect)
 - N400 (condition effect)



Visualization of simulated data

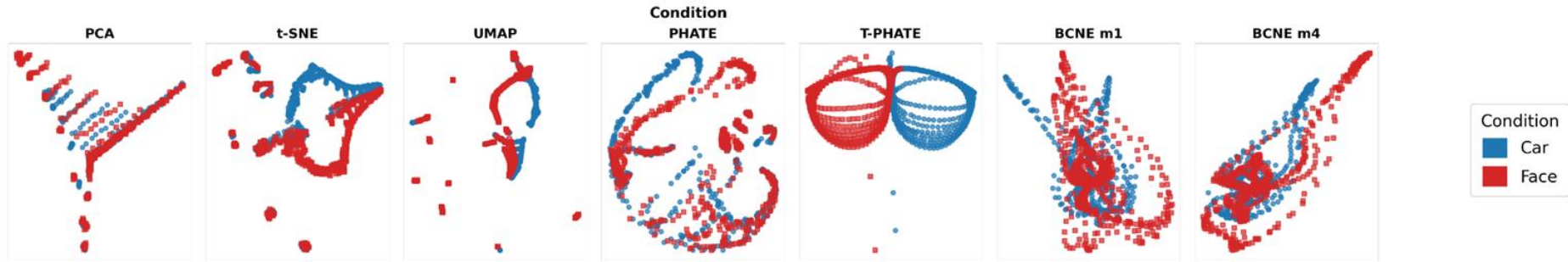
Preprocessing Steps for BCNE

Approach	Input	Condition Separation	Continuous Gradient Separation
Global Average: Average 2000 trials	1 sample	Yes	Yes (fixed projection)
Average by Condition: not feasible	2 samples	Yes	No (averaged away)
Average by Continuous Level: not feasible	10 total (10 levels)	No (averaged away)	Yes
Average by Condition × Continuous: Implemented	20 total (2×10)	Yes	Yes (ordered)
No Averaging (All 2000 trials)	2000 total	Yes	Yes (scattered ordering)

Input and Preprocessing steps

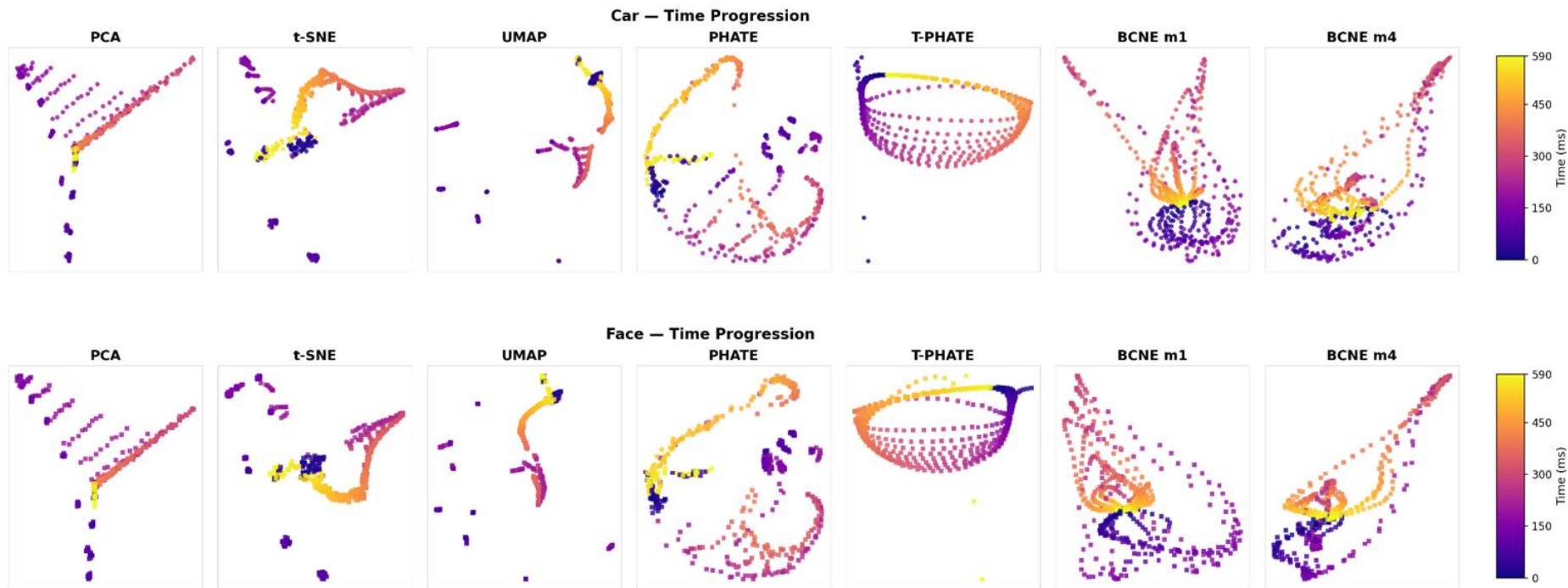
Method	Input	Preprocessing	Shape for Algorithm
PCA, T-SNE, UMAP, PHATE, T-PHATE	Raw trials	• Group by condition and continuous	(1200, 32)
BCNE	Raw trials	• Group by condition and continuous • Time correlation • Spatial channel mapping	(1200, 6, 6, 1)

Results



- Each dot represents an averaged EEG recording at one timepoint where all 32 electrode measurements are collapsed into a single point.

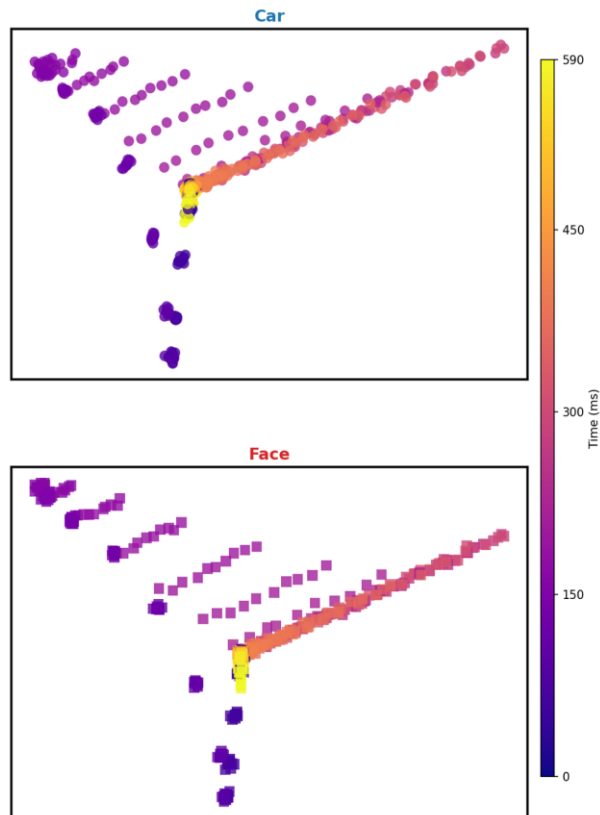
Results



PCA

- PCA can't separate non-linear structure
- Time (0ms → 590ms): Brain activity changes dramatically over time
- Brain activity changes over continuous levels
- No temporal order preserved

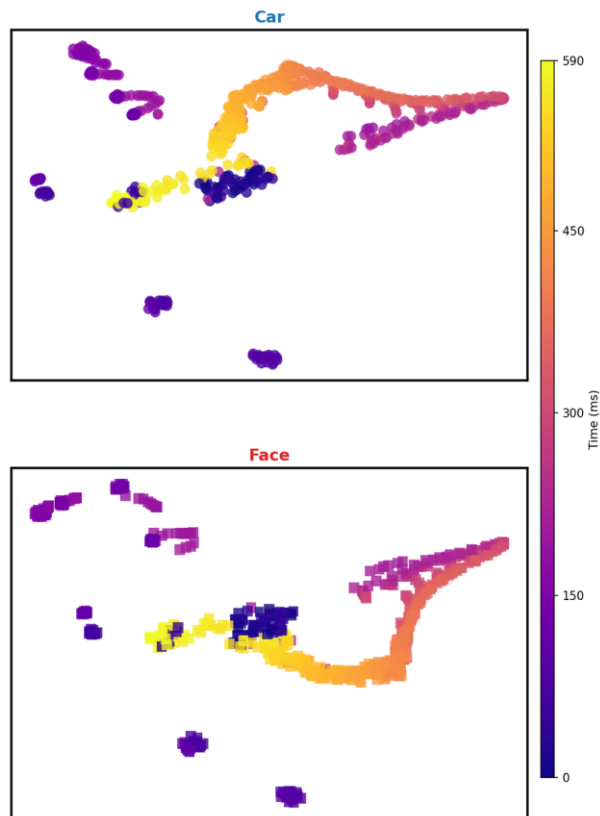
PCA KNN Accuracy (k=8): 0.5942



t-SNE

- Clear separated blobs
- Preserves local neighborhoods defined by the 8 nearest neighbors
- Time points are scattered
- No temporal order preserved
- Good for clustering

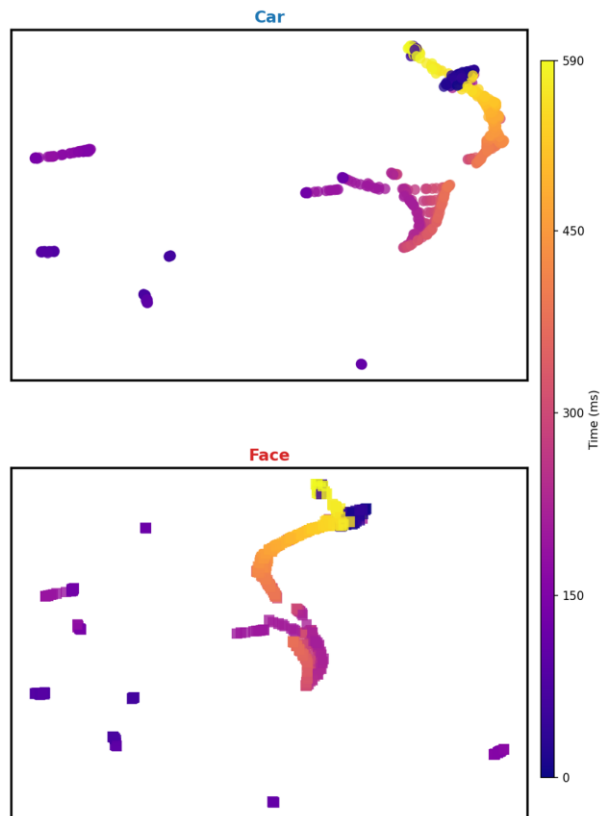
t-SNE KNN Accuracy (k=8): 0.8425



UMAP

- Preserves local and global structure
- Tighter clusters than t-SNE
- Better condition separation
- No smooth time progression

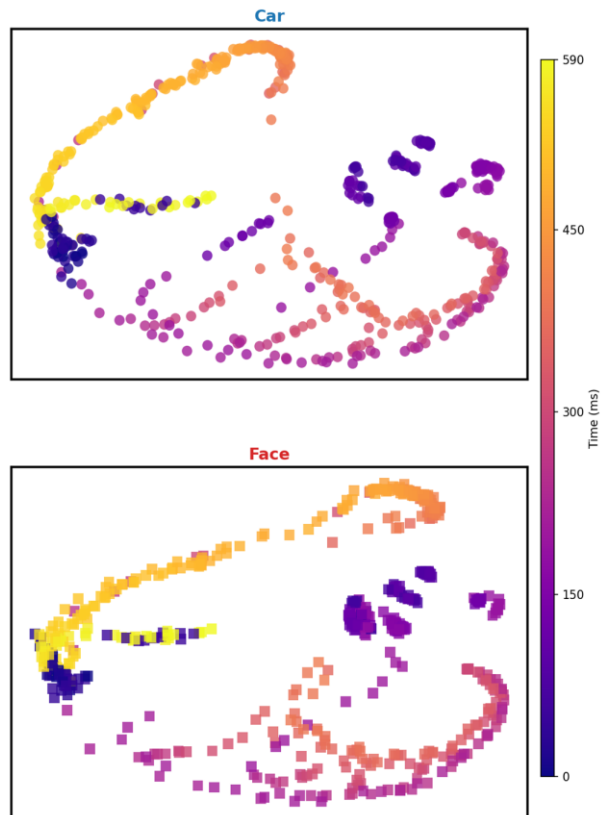
UMAP KNN Accuracy (k=8): 0.8225



PHATE

- Captures geometry/branching structure of data
- Smoother, more trajectory-like shapes
- No temporal order preserved

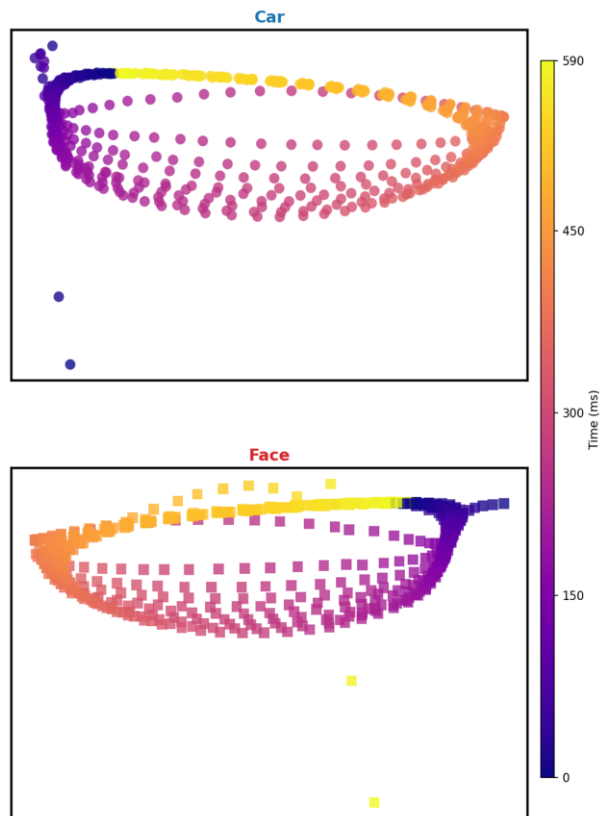
PHATE KNN Accuracy (k=8): 0.7058



TPHATE

- The clearest smooth time gradient flowing from early to late time points
- Conditions following different but coherent paths.
- Temporal Structure Preserved

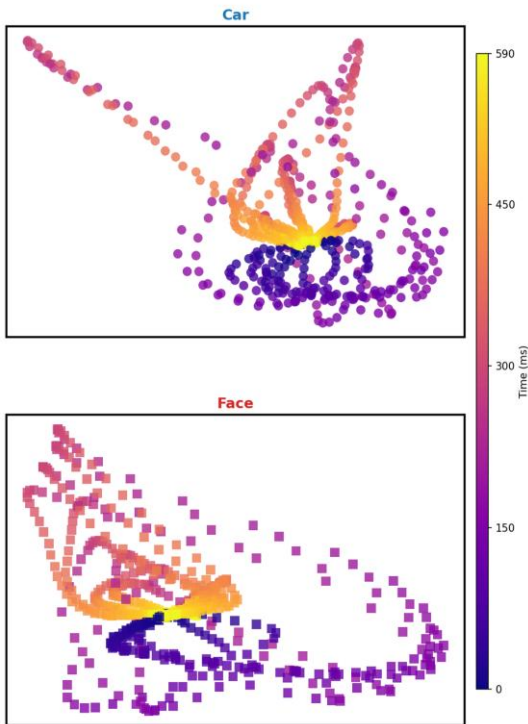
T-PHATE KNN Accuracy (k=8): 0.9742



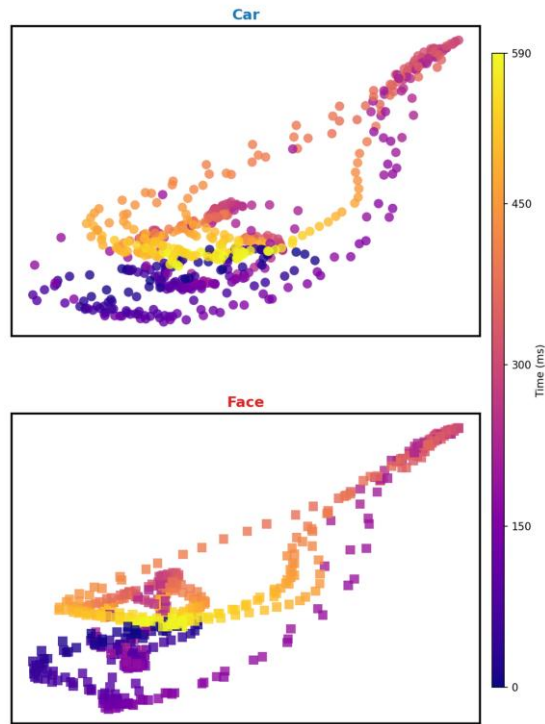
BCNE

- The first recursion primarily captures local neighborhoods and has not yet learned the deeper global structure
- M4 model is better as it **learns** the true structure of the data

BCNE m1 KNN Accuracy (k=8): 0.6058



BCNE m4 KNN Accuracy (k=8): 0.7025



Current problem 1: Identifying training strategy for BCNE

Option 1:

Single global average

Input Data: 1 averaged signal
(60 timepoints $\times$ 32 channels)

Option 2:

20 group averages

Input: 20 groups (100 trials each)
Total: 1200 samples

Current problem 2:

How to verify the structure of embeddings

Qualitative checks:

- Condition Separation: Face vs. Car
- Trajectory Smoothness
- Component Detectability: Are P100, N170, P300, and N400 visible at expected time points?
- Temporal Structure: Is timing preserved?

Quantitative checks:

- K-Nearest Neighbours
- Trustworthiness
- Continuity
- Temporal RSA

Metrics Evaluation

Method	KNN (k=8)	KNN (k=15)	Trustworthiness	Continuity
PCA	0.594	0.594	0.873	0.961
t-SNE	0.842	0.818	0.997	0.994
UMAP	0.823	0.816	0.994	0.993
PHATE	0.706	0.694	0.944	0.983
T-PHATE	0.974	0.974	0.957	0.944
BCNE m1	0.853	0.844	0.875	0.920
BCNE m4	0.833	0.828	0.837	0.904

Timeline

Tasks	Thesis Duration (December 2025- June 2026)							
	December	January	February	March	April	May	June	July
Proposal								
Literature Review								
Introduction presentation								
Data Simulation								
Implementation								
Metric Evaluation								
Midterm presentation								
Analysis								
Final presentation								
Review								

References

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Example of input data for BCNE

